

Akhil Balineni

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PROFESSIONAL SUMMARY

Engineer with Experience in Software Development, specializing in building scalable and high-performance applications. Computer Science Graduate at ASU, seeking a Software Development Engineer role to apply expertise in front-end technologies, API design, and cloud-based infrastructure to develop efficient, responsive, and reliable software solutions

EDUCATION

Master of Science in Computer Science

August 2023 - May 2025

Arizona State University, Tempe, Arizona

3.97 / 4.00

Coursework: Cloud Computing, Data Mining, Data Processing at Scale, Statistical Machine Learning, Data Analytics

Bachelor of Technology in Computer Science & Engineering

July 2019 - June 2023

Sri Ramaswamy Memorial Institute of Science and Technology, India

3.76 / 4.00

Coursework: Artificial Intelligence, Data Structures & Algorithm design, Statistics, Software Engineering, Data Science

WORK EXPERIENCE

Software Development Engineer Intern | The Next Step | USA (Remote)

June 2021 – July 2023

- Developed and optimized backend services for a web-based Service Request Portal using **Java** and **Spring Boot**, enabling seamless creation, tracking, and management of customer requests and tasks
- Developed a **RESTful API** for the notification service, automating email dispatch for user registration and service updates. Optimized email delivery, reducing average processing time to **5 seconds** per email
- Integrated **Apache Kafka** for event-driven communication between microservices, minimizing data processing latency by **40%** and enhancing system scalability by **65%**, ensuring reliability under varying workloads.
- Implemented efficient data handling techniques using **PL/SQL** stored procedures and JDBC routines, streamlining SQL query execution within the **Oracle Database environment**
- Refactored backend architecture, incorporating **lazy loading** for handling **40,000+** records and implementing server-side caching, leading to a **40%** reduction in initial data load time and improved response efficiency

PROJECTS

Elastic Face Recognition System in Videos | AWS Greengrass, EC2, Lambda, S3, SQS, MQTT, MTCNN, Python, Paas, Saas

- Built a distributed face recognition pipeline using **AWS Greengrass** for real-time edge detection (**MTCNN**) and **AWS Lambda** for cloud-based identity classification, enabling sub-second latency and 100% accuracy on video frames
- Integrated **MQTT** and **SQS** for secure, asynchronous communication between edge and cloud layers, processing 200+ concurrent requests efficiently using **EC2**, **S3**, and custom autoscaling logic without Auto Scaling Groups

OCT-Based Retinal Disease Detection Using VGG16 Transfer Learning | Deep Learning, Computer Vision

- Developed an ML model using **VGG16** transfer learning with **ReLU** activation, **Softmax** for classification, and **Adam optimizer** to detect retinal diseases from OCT scans
- Achieved **94.9%** training accuracy with a validation loss of **0.0025**, ensuring precise pre-diagnosis

Yelp Dataset Mining and Business Insights | PySpark, Hadoop, Hive, Python, SQL, Matplotlib

- Designed and executed distributed data processing workflows using **PySpark** on **Hadoop** and **Hive**, writing optimized **SQL queries** to perform complex joins, aggregations, and sentiment-based segmentation over 5M+ Yelp reviews
- Built end-to-end data pipelines for reviewer profiling, integrating Python scripts and **HiveQL** to generate time-series insights and user engagement visualizations across elite and non-elite user cohorts

TECHNICAL SKILLS

- Programming Skills:** Python, Java, C++, C#, C, Shell, Perl
- Web Development:** HTML5, CSS, PHP, JavaScript, Django, Springboot, ReactJS, AngularJS, Node.js, Microservices, .NET
- Tools:** Selenium, Flask, Docker, Hadoop, Spark, PySpark, FastAPI, Android Studio, Matlab, Axure RP, Figma, Tableau
- ML Libraries:** TensorFlow, Scikit-Learn, Keras, PyTorch, NumPy, Pandas, SciPy, Matplotlib
- Technical Proficiencies:** GCP, Object-Oriented, Agile, Ubuntu, Linux, SQL, MySQL, MongoDB, Full Stack Development